Introduction

The World Is a Chessboard

A Strategic Reality

We often think life is a series of random events. **It is not.**[6pt] It is a game, sometimes cooperative, sometimes competitive, but *always strategic*.

- From the vegetable vendor negotiating prices ...
- ... to the diplomat averting war.
- **We are all players.**
- The question is: are you playing with a strategy?

Who This Is For

- Professionals & Managers
- Negotiators & Leaders
- Entrepreneurs & Investors
- Curious Minds
- *Anyone who interacts with other humans*

What Is Game Theory?, A Brief History

Origins

- **1944**, Von Neumann & Morgenstern publish *Theory of Games and Economic Behaviour*
- **1950**, John Nash introduces the *Nash Equilibrium*
- **1994**, Nash, Harsanyi & Selten win the Nobel Prize in Economics
- **2005, 2012**, Further Nobel awards for Mechanism Design and Matching Theory

Key Pioneers

- **John von Neumann**, Founder of modern game theory
- **John Nash**, Equilibrium concept
- **Thomas Schelling**, Focal points & brinkmanship
- **Kahneman & Tversky**, Behavioural economics
- **Kautilya (Chanakya)**, Ancient strategic wisdom

Why Game Theory Matters Today

Traditional Thinking

- Assumes you are in control
- Optimises in isolation
- "How do I get the best outcome?"
- *Like an engineer solving physics*

Game-Theoretic Thinking

- Assumes others are strategic too
- Optimises in interaction
- "How do I get the best outcome **given what others will do?**"
- *Like a chess player anticipating moves*

Why Game Theory Matters Today

Where It Applies

- **Business**, Pricing, competition, M&A
- **Politics**, Elections, coalitions, treaties
- **Personal Life**, Negotiation, relationships
- **Technology**, Platform strategy, auctions
- **Society**, Public goods, voting, norms
- **Finance**, Auctions, trading, insurance

Key Insight

You are **never** playing solitaire. You are always playing chess.

How to Use This Playbook

Structure of Each Concept

1. **Introduction**, What is it?
2. **Situation**, When does it apply?
3. **Steps**, How to use it?
4. **Examples**, Real-world anchors
5. **Takeaway**, The bottom line

Sections Covered

1. Foundational Concepts
2. Game Types & Structures
3. Decision-Making Approaches
4. Credibility & Commitment
5. Strategic Interaction in Relationships
6. Competitive Advantage
7. Information & Uncertainty
8. Negotiation & Bargaining
9. Market Design
10. Collective Challenges
11. Prospect Theory

How to Use This Playbook

A Word of Warning

Game Theory is a **map**, not the territory.[6pt] Real life is messy. People are irrational. Rules change.[6pt] Use these models as lenses, not as algorithms.

The Infinite Game

In the "Finite Game" of a football match, you play to *win*.[4pt] In the "Infinite Game" of life, you play to *keep playing*.[4pt] Understand the incentives. Respect the players.

Foundational Strategic Concepts

Nash Equilibrium: The Stable State

- **Introduction**: A state where no player can improve their outcome by changing their strategy alone. It is the point at which the game *self-locks*, often into a suboptimal outcome for all.
- **Situation**: Arises in any repeated interaction where actors respond to each other, arms races, price wars, social norms, congestion. Whenever you are stuck in a bad pattern everyone perpetuates.
- **Steps**: (1) List all players and their possible strategies. (2) Map payoffs for each combination. (3) For each player, find the best response to every opponent strategy. (4) Identify where best responses align, that is the equilibrium. (5) Ask: is it efficient? If not, what changes the game?
- **Examples**: Everyone stands at a cricket match so no one sees better; arms races leaving all nations poorer; airlines locked in price wars destroying profitability.

Dominant Strategy: The Safe Move

- **Introduction**: A dominant strategy delivers the highest payoff for a player *regardless of what others do*. It is the "no-brainer", you need not read the opponent's mind when you have one.
- **Situation**: Look for dominant strategies before trying to anticipate opponents. Useful in competitive exams, auctions, market pricing, and any situation where one choice outperforms all others unconditionally.
- **Steps**: (1) Draw a payoff matrix. (2) For each of your strategies, find the payoffs against every opponent move. (3) Check if one row always beats the others. (4) If yes, that is your dominant strategy, play it. (5) If not, identify your opponent's dominant strategy and best-respond to it.
- **Examples**: Amazon's relentless cost leadership; index fund investing over stock-picking; vaccination during a pandemic regardless of what neighbours do.

Zero-Sum Games: Winner-Takes-All

- **Introduction**: In a zero-sum game the total available benefit is fixed, one player's gain is another's exact loss. These are games of pure conflict where cooperation is mathematically impossible.
- **Situation**: Applicable where resources are genuinely scarce and non-expandable: elections, IIT seats, market share in a stagnant industry, athletic competitions, inheritance division.
- **Steps**: (1) Verify the game is truly zero-sum before treating it as one. (2) Identify your relative strengths vs. opponents. (3) Use minimax strategy, minimise maximum possible loss. (4) Look for dominant positions to exploit. (5) Consider whether the game can be redesigned as positive-sum.
- **Examples**: Poker tables; election seat counts; dividing a fixed estate; ranking systems where one rise means another's fall.

Positive-Sum Games: Creating Mutual Benefit

- **Introduction**: Positive-sum games allow the total gains to grow, "growing the pie" rather than fighting over slices. Trade, collaboration, and innovation are classic examples where mutual gain is possible.
- **Situation**: Most business partnerships, trade agreements, marriages, and open-source projects. When both parties gain from interaction, the strategic goal shifts from defeating to creating and coordinating.
- **Steps**: (1) Map potential joint gains from cooperation. (2) Identify each party's comparative advantage. (3) Design a split of gains that both prefer over non-cooperation. (4) Build trust mechanisms to prevent defection. (5) Iterate, reputation and repeat interaction sustain positive-sum outcomes.
- **Examples**: Bangalore's tech ecosystem where new firms benefit incumbents; open-source software; international trade agreements; ridesharing platforms benefiting both drivers and riders.

Game Types and Structures

Prisoner's Dilemma: The Trust Problem

- **Introduction**: The most famous paradox in game theory, individual rationality (self-protection) leads to collective irrationality (both suffer). Mutual cooperation would be best, but each defects, fearing betrayal.
- **Situation**: Any situation where independent self-interest produces a worse collective outcome: price wars, arms races, doping in sports, environmental pollution, negative political campaigning.
- **Steps**: (1) Recognise the dilemma structure in your situation. (2) Assess the shadow of the future, is there a next interaction? (3) Build communication and trust mechanisms. (4) Introduce contracts or penalties for defection. (5) Use tit-for-tat strategies in repeated play.
- **Examples**: Two sweet shops undercutting each other to thin margins; competing nations in arms races; athletes using performance-enhancing drugs; public littering.

Chicken Game: The Dangerous Standoff

- **Introduction**: A test of nerves where the worst outcome is mutual destruction. Unlike the Prisoner's Dilemma, someone *must* yield. The winner is the one who convincingly cannot back down.
- **Situation**: High-stakes confrontations where one party must blink: geopolitical crises, union vs. management standoffs, legislative gridlock, corporate takeover battles, sovereign debt negotiations.
- **Steps**: (1) Assess the cost of mutual destruction to both sides. (2) Signal commitment credibly, make retreat costly or visible. (3) Look for face-saving offramps for the opponent. (4) Avoid symmetrical escalation. (5) Know your red lines before the confrontation begins.
- **Examples**: Cuban Missile Crisis; union strikes vs. management lockouts; government debt-ceiling standoffs; two buses on a narrow road.

Stag Hunt: Coordination Through Trust

- **Introduction**: A coordination game with two equilibria, a superior cooperative outcome (the Stag) and an inferior but safe solo outcome (the Hare). The challenge is not conflict but *assurance*: knowing others will commit.
- **Situation**: Climate accords, new technology standards, currency adoption, startup equity decisions, team sports defence, any endeavour where joint success requires full participation.
- **Steps**: (1) Communicate the payoff difference between cooperation and going solo. (2) Build pre-commitment mechanisms. (3) Start with a small, credible coalition. (4) Provide insurance against defection. (5) Create repeated-game conditions so defectors can be punished.
- **Examples**: Nations adopting climate accords; tech industry standardising on USB-C; startups choosing equity over salary; dam-building village cooperatives.

Battle of the Sexes: Preference Conflicts

- **Introduction**: Both players want to coordinate but disagree on *which* coordination point to choose. It is not pure conflict, both want togetherness, but power determines whose preference wins.
- **Situation**: Merger decisions (who becomes CEO?), standard-setting wars (Mac vs PC), coalition government formation, meeting location choices, holiday planning.
- **Steps**: (1) Confirm both parties genuinely prefer coordination to no deal. (2) Identify the current default, who benefits from the status quo? (3) Use side-payments or alternating concessions ("I'll watch cricket now, you choose next time"). (4) Make concessions visible and reciprocal. (5) Formalise agreements to avoid renegotiation.
- **Examples**: Merger CEO selection; choosing a vacation destination; technology standard wars; political coalition ministry allocation.

Ultimatum Game: Fairness Over Profit

- **Introduction**: People will sacrifice their own gain to punish perceived unfairness. This game demonstrates that humans are emotional moralists, not purely rational profit-maximisers, destroying the "Rational Man" economic model.
- **Situation**: Any offer/rejection setting: wage negotiations, settlement offers, supplier contracts, brand boycotts. Whenever perceived fairness can override strict economic logic.
- **Steps**: (1) Frame offers so they appear fair, not just profitable to you. (2) Use anchoring to set fairness expectations early. (3) Give the other party a sense of agency in the process. (4) Avoid visibly large asymmetries in payoffs. (5) Expect rejection of "logically reasonable" offers that feel insulting.
- **Examples**: Labour strikes rejecting technically positive wage offers; restaurant tipping even when you will never return; brand boycotts over perceived exploitation.

Public Goods Game: The Free-Rider Problem

- **Introduction**: A public good is non-excludable and non-rivalrous. The free-rider problem emerges when individuals consume more than their fair share while paying less than their fair share, destroying the commons.
- **Situation**: Wikipedia donations, national defence, clean air, office coffee funds, open-source software maintenance, and voluntary neighbourhood upkeep.
- **Steps**: (1) Identify whether the good is genuinely non-excludable. (2) Introduce social norms or visibility (people contribute more when observed). (3) Use conditional cooperation mechanisms ("I give if others give"). (4) Create excludability where possible (paywalls, memberships). (5) Regulate via taxation or mandated contribution.
- **Examples**: Wikipedia donation drives; national vaccination campaigns; office kitchen cleanliness; funding public broadcasting.

Matching Pennies: Pure Unpredictability

- **Introduction**: A purely competitive zero-sum game with no stable pure strategy. Your only winning approach is randomisation, if you have a pattern, the opponent exploits it.
- **Situation**: Penalty kicks; tax audit selection; security screening; military patrol routes; marketing launch timing against competitors, anywhere your adversary benefits from predicting you.
- **Steps**: (1) Identify that you are in a purely competitive, no-equilibrium-in-pure-strategies game. (2) Calculate the optimal mixed strategy probabilities. (3) Introduce genuine randomisation (dice, random number generators). (4) Resist the temptation to use "gut feel" patterns. (5) Monitor whether the opponent is exploiting any pattern and adjust.
- **Examples**: Penalty kick direction; customs inspection selection; submarine evasion routes; Rock-Paper-Scissors.

Cooperative Game Theory: Sharing the Gains

- **Introduction**: Cooperative game theory asks: when players can form coalitions and make binding agreements, how should gains be fairly and stably divided? The Shapley Value and the Core are the key solution concepts.
- **Situation**: Startup profit-sharing, political coalition formation, airline alliances, pharmaceutical joint ventures, supply-chain cost allocation, international carbon agreements.
- **Steps**: (1) Identify all possible coalitions and their total values. (2) Calculate each player's marginal contribution (Shapley Value). (3) Check if the allocation is in the Core, no sub-coalition wants to break away. (4) Negotiate side-payments if the allocation is outside the core. (5) Build binding agreements and enforcement mechanisms.
- **Examples**: Law firm profit splits; coalition government ministry allocations; airline revenue-sharing alliances; joint-venture R&D cost-sharing.

Decision-Making Approaches

Backward Induction: Think Backwards, Act Forward

- **Introduction**: To solve a sequential game, examine the final stage, find the optimal move there, then roll back logic to the start. It prevents moves today that create traps tomorrow.
- **Situation**: Chess endgames, business exit strategy design, career planning, negotiations with defined endpoints, syllabus design, retirement planning.
- **Steps**: (1) Define the final desired outcome clearly. (2) Work backwards: what must be true one step before the end? (3) Continue rolling back, mapping each decision node. (4) Identify the first move that leads to the desired end. (5) Check for credibility, are all intermediate steps actually incentive-compatible?
- **Examples**: Chess endgame analysis; Indian parents planning children's education backwards from career goal; startup exit strategy mapped from IPO back to MVP.

Forward Induction: Reading Hidden Intentions

- **Introduction**: Past actions reveal future intent. If a player makes a move that would be irrational *unless* they held a specific strong position, you can infer that strong position.
- **Situation**: Interpreting competitor signals in markets, reading investor confidence from founder behaviour, recruiting and interviewing, diplomacy, strategic partnership evaluation.
- **Steps**: (1) Observe a costly or unusual action by the other party. (2) Ask: what beliefs would make this action rational? (3) Update your assessment of their type or intention. (4) Adjust your strategy accordingly. (5) Use the same logic to send credible signals about your own type.
- **Examples**: A founder taking a Rs. 1 salary signals supreme confidence to investors; large Super Bowl ad spend signals financial strength; university degrees signal perseverance.

Sequential Games: Turn-Based Strategy

- **Introduction**: In sequential games, players move one after another and can observe prior moves. Time and order of moves determine power, first movers set the tone, last movers often control the outcome.
- **Situation**: Real estate negotiation, chess, lobbying, wage bargaining, courtroom strategy, Stackelberg competition in duopoly markets.
- **Steps**: (1) Map the game tree with all decision nodes. (2) Apply backward induction. (3) Identify who benefits from moving first vs. last. (4) Consider pre-commitment to shape subsequent play. (5) Use observable actions to constrain the opponent's future choices.
- **Examples**: House purchase negotiation (offer → counter → walk away → call back); Stackelberg leader in output competition; chess opening theory.

Simultaneous Games: Deciding Under Uncertainty

- **Introduction**: Players act at the same time without observing each other's choices. Strategy must be based on probability and beliefs about opponent rationality, not observed moves.
- **Situation**: Sealed tender auctions, stock market investing, general elections, simultaneous product launches, penalty kicks where the decision is truly simultaneous.
- **Steps**: (1) Enumerate all strategies and build a payoff matrix. (2) Search for dominant strategies. (3) If none exist, find Nash Equilibrium in pure strategies. (4) If none in pure strategies, compute mixed strategy Nash Equilibrium. (5) Use prior information about opponent types to refine predictions.
- **Examples**: Sealed government contract bids; buying stocks before earnings announcements; Rock-Paper-Scissors.

Subgame Perfect Equilibrium: Credible Plans

- **Introduction**: Nash equilibrium can include non-credible threats that are never tested. Subgame perfection demands that your strategy must be optimal at *every* decision point, even in branches of the game tree you hope never to reach.
- **Situation**: Labor negotiations, military deterrence, corporate takeover defence, parent-child discipline, contract enforcement, anywhere threats or promises must be believed.
- **Steps**: (1) Draw the full game tree. (2) Apply backward induction at every node. (3) Remove strategies that are irrational at any sub-game. (4) Ask: if this branch were reached, would I actually follow through? (5) If not, redesign your commitment to make it credible.
- **Examples**: Parent who always carries through on stated consequences; proportional military response doctrines; poison-pill takeover defences that are actually executable.

Minimax Strategy: Minimising Maximum Loss

- **Introduction**: A conservative strategy that looks at the worst-case scenario of every decision and selects the one with the least-bad maximum loss. It is playing not to lose rather than playing to win.
- **Situation**: Risk management, insurance, cybersecurity, crisis planning, healthcare protocols, wherever catastrophic downside must be avoided even at cost of upside.
- **Steps**: (1) List all your strategies and all opponent strategies. (2) Find the worst payoff for each of your strategies (the "minimum"). (3) Select the strategy with the highest minimum payoff. (4) Compare minimax cost against expected value analysis. (5) Adjust based on risk tolerance and reversibility of outcomes.
- **Examples**: Travel insurance; diversified investment portfolios; pre-nuptial agreements; cybersecurity breach response planning.

Bayesian Games: Decisions with Incomplete Information

- **Introduction**: Most real-world games involve uncertainty about opponents' types, payoffs, or capabilities. Players form beliefs, observe actions, update via Bayes' rule, and choose strategies maximising expected payoffs given current beliefs.
- **Situation**: Used car markets, job interviews, insurance underwriting, military intelligence assessment, product demand forecasting, dating and relationship formation.
- **Steps**: (1) Assign prior probability distributions over opponent types. (2) Identify what signals or actions reveal information. (3) Update beliefs using Bayes' Rule as evidence arrives. (4) Compute expected payoffs across opponent types. (5) Choose strategy maximising expected payoff; recognise your actions also signal your type.
- **Examples**: Employer updating beliefs about candidate quality through interview; used-car buyer reading seller signals; poker player updating hand estimate from betting patterns.

Credibility and Commitment

Strategic Commitment: Limiting Your Own Options

- **Introduction**: Paradoxically, you can increase bargaining power by *reducing* your own options. If you credibly cannot back down, the other party must adapt to your position.
- **Situation**: Wage negotiations, competitive market entry threats, diplomatic standoffs, self-control problems, anywhere restricting future flexibility strengthens your current position.
- **Steps**: (1) Identify the decision you want the other party to take. (2) Find a way to make your own retreat genuinely costly or impossible. (3) Make the commitment visible and verifiable. (4) Ensure the commitment is specific, not vague. (5) Time the commitment before negotiations begin.
- **Examples**: Cortés burning his ships; automatic pension deductions; publicly announced deadlines; contracts with heavy penalty clauses; hunger strikes.

Pre-Commitment Devices: Binding Your Future Self

- **Introduction**: We are often two people, the planner and the impulsive doer. Pre-commitment devices are strategies the planner uses to constrain the doer before temptation strikes.
- **Situation**: Dieting, saving, avoiding addictions, maintaining study habits, managing social media overuse, any area where your future self's impulsive decisions conflict with your present-self's goals.
- **Steps**: (1) Identify the specific moment of temptation. (2) Introduce a barrier before that moment (not after). (3) Make the barrier automatic, not dependent on future willpower. (4) Increase the cost of overriding the commitment. (5) Use social accountability to amplify the pre-commitment.
- **Examples**: Odysseus tying himself to the mast; deleting social media apps; freezing credit cards in ice; gym buddies; automatic savings transfers.

Credible Threats: Making Promises Believable

- **Introduction**: A threat is only useful when it is credible. If carrying out the threat hurts you more than the opponent, it will be called as a bluff. The incentive structure must make follow-through rational.
- **Situation**: Parental discipline, deterrence strategy, price-match guarantees, union strike threats, contractual penalty clauses, wherever compliance requires the other party to believe punishment is real.
- **Steps**: (1) Ask honestly: would I actually carry this out? (2) Adjust the threat so it costs you less than compliance costs the opponent. (3) Make the threat automatic (remove discretion). (4) Build a track record of following through. (5) Communicate the threat early and unambiguously.
- **Examples**: Nuclear deterrence structures; price-match guarantees (credible threat to rivals); central bank inflation targets; unconditional resignation letters.

Brinkmanship: Playing with Fire

- **Introduction**: Deliberately pushing a situation to the edge of mutual disaster, or creating the appearance of losing control, to frighten the opponent into conceding. The shared risk becomes the leverage.
- **Situation**: Debt-ceiling negotiations, strikes timed to peak season, geopolitical confrontations, diplomatic ultimatums, late-stage acquisition negotiations.
- **Steps**: (1) Assess the cost of mutual disaster, it must be roughly equal or worse for the opponent. (2) Create visible risk of escalation (not just the threat). (3) Maintain plausible deniability, "I may not be able to stop this." (4) Keep an offramp for the opponent to retreat without total humiliation. (5) Know precisely when to de-escalate.
- **Examples**: Indo-Pak border tensions; union strikes during harvest season; debt-ceiling confrontations; Cold War nuclear postures.

Burning Bridges: Eliminating Retreat

- **Introduction**: By deliberately eliminating your own escape routes, you signal unshakeable commitment. When retreat is impossible, threats become credible, bluffs become real, and opponents must take your position seriously.
- **Situation**: Market entry with irreversible investment, public goal announcements, constitutional lock-ins, military positions, start-up pivots where the old product is shut down.
- **Steps**: (1) Identify what escape routes the opponent expects you to take. (2) Eliminate those routes publicly and irreversibly. (3) Signal the elimination to your opponent clearly. (4) Ensure the commitment is specific enough to be believed. (5) Carefully weigh: are you certain enough to make retreat impossible?
- **Examples**: Cortés burning ships in Mexico; no-refund advance-payment policies; publicly announced deadlines; constitutional supermajority requirements; military forces holding indefensible positions.

Reputation Effects: Your History Precedes You

- **Introduction**: In repeated interactions, your past actions determine your future payoffs. A reputation for being "tough" or "honest" changes how others play against you, it is a capital asset that earns compound returns.
- **Situation**: Brand management, credit markets, diplomatic relations, business partnerships, personal professional networks, any context with repeated interaction and observable history.
- **Steps**: (1) Identify what reputation you want to build. (2) Make short-term sacrifices to establish credibility. (3) Ensure your actions are observable. (4) Maintain consistency, reputation is destroyed by exceptions. (5) Invest in platforms where reputation is recorded (ratings, testimonials, public records).
- **Examples**: Tata Group's century-long reputation; credit scores; eBay seller ratings; central bank inflation credibility; Mafia omertà.

Escalation of Commitment: The Sunk Cost Trap

- **Introduction**: The irrational decision to continue a failing course of action because of past investments. Rational strategy demands looking only forward, past costs are gone and irrelevant to future decisions.
- **Situation**: Ongoing projects with poor ROI, failing relationships, bad stocks held too long, unfinished degrees, prolonged wars, any situation where sunk cost drives continued investment.
- **Steps**: (1) Separate past costs from future costs and benefits. (2) Ask: if I were starting today, would I make this investment? (3) Establish clear kill-criteria in advance. (4) Assign an independent reviewer not emotionally attached to the project. (5) Reframe exit as strategic intelligence, not failure.
- **Examples**: The Vietnam War; waiting for a bus already 30 minutes late; gambling to "win it back"; keeping a failing business unit alive.

Strategic Interaction in Relationships

Tit-for-Tat: Reciprocity Works

- **Introduction**: The winning strategy in repeated Prisoner's Dilemma tournaments: cooperate first, then mirror whatever the opponent did last round. Simple, forgiving, but retaliatory, the algorithm of human morality.
- **Situation**: Trade negotiations, business partnerships, gifting cultures, arms control agreements, neighbourhood relations, any long-term repeated interaction where defection must be deterred.
- **Steps**: (1) Open by cooperating unconditionally. (2) Immediately mirror any defection with one round of defection. (3) Return to cooperation the moment the opponent cooperates. (4) Never defect first. (5) Communicate the strategy so the opponent understands the pattern.
- **Examples**: Trade tariff reciprocity; reciprocal dinner invitations; ceasefire agreements conditioned on compliance; eye-for-an-eye justice systems.

Shadow of the Future: Long-Term Thinking

- **Introduction**: Cooperation is sustainable only when the "shadow of the future" looms large. If interactions repeat indefinitely, the long-term reputational loss from cheating outweighs the short-term gain.
- **Situation**: Family businesses, small-town communities, subscription models, diplomatic relations, supplier relationships, wherever continuity of interaction creates accountability.
- **Steps**: (1) Increase the probability and salience of future interactions. (2) Reduce the discount rate on future payoffs. (3) Establish visible accountability systems. (4) Build reputational stakes into short-term interactions. (5) Create switching costs that make exit from the relationship costly.
- **Examples**: Not cheating your local vendor because you see them daily; subscription loyalty programs; diplomatic relations sustained by future needs; small-town honesty norms.

Signalling: Sending Messages Through Action

- **Introduction**: When one party holds private information, they must use observable actions to convey it credibly. A good signal must be both observable and difficult to fake, otherwise it conveys nothing.
- **Situation**: Job market signalling, brand advertising, corporate dividend policy, product warranties, personal reputation management, any information-asymmetric interaction.
- **Steps**: (1) Identify the hidden information you need to convey. (2) Find an action that is cheap for you (high type) but expensive for a low type to mimic. (3) Make the signal observable. (4) Maintain signal consistency over time. (5) Verify that recipients correctly interpret the signal.
- **Examples**: Wedding rings as availability signals; dividends signalling corporate financial health; peacock tails as evolutionary fitness signals; product warranties signalling quality.

Costly Signalling: Credibility Through Sacrifice

- **Introduction**: Talk is cheap; for a signal to be trusted in high-stakes games it must be too expensive for a "faker" to afford. The cost *is* the proof.
- **Situation**: High-stakes relationship formation, luxury brand signalling, investor confidence building, gang initiation rites, religious fasting, any context where cheating would be profitable unless signalling is costly.
- **Steps**: (1) Identify what claim you need others to believe. (2) Find a costly action that only someone with the genuine claim would rationally take. (3) Ensure the cost is observable and verified. (4) Calibrate cost: high enough to deter fakers, not so high it deters honest signalers. (5) Repeat over time to build a track record.
- **Examples**: Expensive engagement rings; charitable donations by the wealthy; fraternity initiation costs; luxury goods as wealth signals; religious fasting as community commitment.

Rational Irrationality: Strategy Through Unpredictability

- **Introduction**: Being perfectly rational makes you predictable and exploitable. Feigning, or genuinely being, irrational alters the opponent's payoff matrix, forcing them to yield to avoid an unpredictable catastrophe.
- **Situation**: Geopolitical standoffs, extreme negotiation leverage, crowd control, deterring bullying, anywhere the opponent's certainty about your behaviour is a weapon against you.
- **Steps**: (1) Identify where your predictability is being exploited. (2) Introduce deliberate unpredictability or "madman" signals. (3) Ensure the irrationality is credible, genuine emotional investment helps. (4) Have a private off-switch you control even while appearing to lack one. (5) Return to rational behaviour once the leverage has worked.
- **Examples**: Nixon's "Madman Theory" in Vietnam; protester self-immolation as commitment device; doomsday devices in deterrence; road rage escalating beyond rational limits.

Competitive Advantage and Market Strategy

First-Mover Advantage: Being Early Pays

- **Introduction**: The first player to enter a market can lock in resources, customers, and standards. They set the rules that followers must accept. The brand can become the category name itself.
- **Situation**: New technology categories, emerging markets, standard-setting battles, patent races, geographic expansion into underpenetrated regions.
- **Steps**: (1) Identify the untapped market and timing window. (2) Move decisively before the window closes. (3) Use early position to lock in key resources (talent, patents, partners). (4) Set the standard others must comply with. (5) Invest early in switching costs and network effects.
- **Examples**: Xerox defining "photocopying"; Amazon in e-commerce; Coca-Cola's global brand position; Bitcoin as first cryptocurrency; squatters' rights laws.

Second-Mover Advantage: Learning from Mistakes

- **Introduction**: Being first is risky and expensive. Second movers free-ride on the pioneer's investments, learn from their errors, and adopt superior technology once the market is de-risked.
- **Situation**: Technology markets where innovation moves rapidly, emerging product categories after initial market education, regulatory environments that reward fast followers over pioneers.
- **Steps**: (1) Monitor pioneers carefully; let them educate the market. (2) Identify pioneer weaknesses and customer dissatisfactions. (3) Build a superior product on validated demand. (4) Strike at the moment of pioneer fatigue or overextension. (5) Use scale and capital efficiency to leapfrog.
- **Examples**: Google after Yahoo; Facebook after MySpace; Samsung after Apple; generic drugs after patent expiry; Zara replicating runway fashion rapidly.

Entry Deterrence: Protecting Market Territory

- **Introduction**: Incumbents can make it structurally unprofitable for new players to enter by over-investing in capacity, pricing aggressively, or controlling key complements, signalling "don't bother."
- **Situation**: Market leaders facing potential competition, platform companies with network effects, utilities and infrastructure businesses, industries with high capital requirements.
- **Steps**: (1) Signal excess capacity credibly to deter entry. (2) Use loyalty programs and switching costs to lock in customers. (3) Acquire critical input suppliers or distribution channels. (4) Use limit pricing to keep post-entry profits unattractive. (5) File strategic patents around core technology.
- **Examples**: Jio's free-data launch; frequent-flyer mile programmes; patent thickets in pharmaceuticals; exclusive distributor contracts; Amazon's low-margin saturation strategy.

Limit Pricing: Discouraging Competition Through Price

- **Introduction**: Setting prices lower than monopoly level, but above competitive cost, specifically to make market entry unprofitable for potential rivals without sacrificing all incumbent profits.
- **Situation**: Dominant firms in markets with moderate entry barriers, commodity markets with cost-advantage incumbents, digital platforms with significant scale economics.
- **Steps**: (1) Estimate the entrant's cost structure. (2) Set price just below the level that makes entry profitable for them. (3) Ensure price is above your own average cost. (4) Signal that price will stay low even post-entry. (5) Back the signal with visible cost advantages.
- **Examples**: OPEC pricing decisions; Microsoft Windows pricing in the 1990s; India's telecom data pricing wars; Amazon Basics undercutting third-party sellers.

Strategic Alliances: Temporary Partnerships

- **Introduction**: Cooperation between competitors, the "frenemy" relationship, to achieve specific goals like scale or access. Partners cooperate on the platform while competing on the product.
- **Situation**: Airline codeshare agreements, technology standards bodies, joint R&D ventures, political coalitions, supply-chain partnerships with horizontal competitors.
- **Steps**: (1) Identify the specific shared objective that justifies cooperation. (2) Bound the alliance scope, cooperate only where interests align. (3) Establish clear governance and exit mechanisms. (4) Protect proprietary information carefully. (5) Plan for the alliance ending when objectives diverge.
- **Examples**: NATO; airline codeshare agreements; Spotify and Uber; Apple using Samsung screens; pharmaceutical joint ventures for R&D.

Cheap Talk: Words Without Bite

- **Introduction**: Costless communication that does not directly affect payoffs. Because it is free, it is often misleading. But when interests align, cheap talk can successfully coordinate without commitment.
- **Situation**: Dating app profiles, political campaign promises, CEO letters, advertising claims, diplomatic statements, anywhere words are free and verification is costly.
- **Steps**: (1) Distinguish cheap talk from costly signalling. (2) Ask: does the speaker have incentives to lie? If yes, discount heavily. (3) Look for corroborating actions, not just words. (4) Use cheap talk to coordinate when interests align. (5) Design systems that make lying costly to convert cheap talk to credible talk.
- **Examples**: Street vendor's "final offer"; political manifesto promises; CEO shareholder letters; football pre-match trash talk.

Information and Uncertainty

Perfect vs. Imperfect Information

- **Introduction**: Perfect information means all players see all moves and the full game state (chess). Imperfect information means some moves are hidden (poker). This single distinction transforms strategy from calculation to psychology.
- **Situation**: Distinguishing which type of game you are in determines whether to invest in deeper analysis vs. psychological profiling and information management.
- **Steps**: (1) Determine what information is observable vs. hidden. (2) In perfect information games, use backward induction and deep calculation. (3) In imperfect information games, model opponent types and beliefs. (4) Invest in revealing opponent information through screening. (5) Conceal your own information strategically.
- **Examples**: Chess (perfect) vs. Poker (imperfect); transparent vs. sealed-bid auctions; open parliamentary debates vs. backroom diplomacy.

Common Knowledge: Synchronised Understanding

- **Introduction**: Mutual knowledge means everyone knows X. Common knowledge means everyone knows that everyone knows X, to infinite recursion. This transition from private belief to shared awareness enables coordinated mass action.
- **Situation**: Social movements, bank runs, market bubbles, cultural paradigm shifts, anywhere private individual beliefs must become public to trigger collective behaviour change.
- **Steps**: (1) Identify information that is privately held but widely shared. (2) Find a public event that makes the private knowledge common. (3) Anticipate the coordinated behaviour change that follows. (4) Use or create common knowledge to coordinate allies. (5) Suppress common knowledge formation to prevent unwanted coordination.
- **Examples**: Arab Spring via social media; bank runs on shared panic; #MeToo making isolated experiences collective; market bubbles bursting on shared doubt.

Information Asymmetry: The Knowledge Gap

- **Introduction**: When one party has more or better information than the other, the informed party can exploit the gap. This imbalance causes market failures as uninformed parties fear being cheated and exit.
- **Situation**: Doctor-patient, mechanic-driver, real estate agent-buyer, insurer-insured, employer-employee relationships, any principal-agent context with unequal information.
- **Steps**: (1) Identify who holds the information advantage. (2) If uninformed, seek signals, warranties, or third-party verification. (3) If informed, decide whether to disclose (building trust) or withhold (exploiting gap). (4) Design mechanisms that align the informed party's incentives with disclosure. (5) Reduce asymmetry through transparency norms or regulation.
- **Examples**: Mechanics exploiting car-ignorant customers; insider trading; real estate agents prioritising commission over client interest.

Adverse Selection: Hidden Types Cause Problems

- **Introduction**: When quality is hidden, bad options (lemons) drive out good ones. Markets collapse because high-quality types leave rather than subsidise low-quality types, leaving only the worst in the pool.
- **Situation**: Insurance markets, used-car markets, online dating, volunteer armies, all-you-can-eat pricing, any market where participants self-select based on private information.
- **Steps**: (1) Identify the hidden characteristic causing selection. (2) Design screening mechanisms to sort types. (3) Use signalling to allow high types to distinguish themselves. (4) Price discriminate based on revealed preference. (5) Introduce mandatory pooling (e.g., compulsory insurance) to prevent market unravelling.
- **Examples**: Akerlof's used-car "lemons" market; health insurance death spirals; all-you-can-eat buffets attracting the most voracious eaters.

Moral Hazard: Risk Without Responsibility

- **Introduction**: When a person is insulated from the consequences of their actions, they behave more recklessly than if they bore the full risk. Shielding from downside destroys incentives for care.
- **Situation**: "Too Big to Fail" banks, comprehensively insured drivers, rent-controlled tenants, CEOs with golden parachutes, any principal-agent setting where consequences are borne by a third party.
- **Steps**: (1) Identify who bears the cost of risky behaviour. (2) Introduce co-payments, deductibles, or clawbacks to re-align incentives. (3) Increase monitoring to observe hidden actions. (4) Design contracts with performance-based compensation. (5) Use reputational accountability to supplement formal incentives.
- **Examples**: Banks gambling with taxpayer-backed deposits; drivers with comprehensive coverage taking more risks; CEOs with severance packages taking reckless bets.

Screening: Testing to Reveal Type

- **Introduction**: The uninformed party sets up a test that induces the informed party to reveal hidden information through self-selection. The design makes lying costly, so truth-telling becomes rational.
- **Situation**: Job interviews, insurance underwriting, university admissions, credit scoring, deductible-based insurance pricing, wherever the uninformed party needs to sort by hidden quality.
- **Steps**: (1) Identify the hidden type that matters (health, ability, commitment). (2) Design a menu of options where each type prefers the option intended for them. (3) Ensure the "wrong" choice is costly enough to deter mimicry. (4) Observe self-selection and update beliefs. (5) Refine the screen as participants learn to game it.
- **Examples**: Insurance deductibles revealing health risk; Saturday-night stay airline pricing separating business from leisure; coding tests in job interviews.

Negotiation and Bargaining

Bargaining Power: Influencing Expectations

- **Introduction**: Your power in any negotiation is determined not by your loudness but by your BATNA, Best Alternative to a Negotiated Agreement. The party who can most credibly walk away controls the table.
- **Situation**: Salary negotiations, merger talks, hostage negotiations, real estate deals, supplier contracts, union collective bargaining, all negotiations where alternatives determine leverage.
- **Steps**: (1) Identify and actively improve your BATNA before negotiating. (2) Try to understand and weaken the other party's BATNA. (3) Never reveal a weak BATNA. (4) Signal your BATNA credibly through actions, not just words. (5) Anchor the negotiation range early with your opening offer.
- **Examples**: Having another job offer in salary negotiations; water vendor in a desert vs. one by a river; buyers in a buyer's market; monopolists with no alternatives facing them.

Nash Bargaining Solution: Fair Division

- **Introduction**: A mathematical approach to finding a "fair" bargaining outcome satisfying efficiency (nothing wasted), symmetry (equal bargainers get equal shares), and individual rationality. It creates a focal point for agreement.
- **Situation**: M&A pricing, climate burden-sharing agreements, royalty splits, plea bargaining, joint-venture profit allocation, wherever a principled division of surplus is needed.
- **Steps**: (1) Identify each party's outside option (disagreement payoff). (2) Calculate the total surplus from cooperation. (3) Find the split that maximises the product of surplus gains for both parties. (4) Check for symmetry adjustments based on relative patience or risk aversion. (5) Use the solution as a benchmark to evaluate proposed deals.
- **Examples**: M&A deal pricing relative to stand-alone values; climate agreement burden-sharing; management-union wage settlements avoiding strikes.

Rubinstein Bargaining Model: Patience Wins

- **Introduction**: When two parties alternate offers over time, each delay costs both sides as the pie shrinks. But the more patient party captures a larger share. With rational players, the first offer is accepted immediately, delay signals irrational commitment or private information.
- **Situation**: Real estate negotiations with carrying costs, labour strikes, divorce proceedings, acquisition talks with dwindling startup runway, international trade negotiations.
- **Steps**: (1) Assess your discount rate, how much does delay cost you? (2) Assess the opponent's discount rate, do they have burning runway? (3) Make your first offer close to the equilibrium solution (it should be accepted). (4) Signal patience credibly. (5) Recognise that real delays indicate incomplete information or irrational commitment, adjust accordingly.
- **Examples**: Car price negotiation as both parties' time costs mount; union strikes where both sides bleed daily; startup acquisition as funding runs out.

Rent-Seeking: Gaining Without Creating

- **Introduction**: Spending resources to increase one's share of existing wealth without creating new wealth. It is economically wasteful, consumes talent in zero-sum activities, and corrupts strategic incentives.
- **Situation**: Lobbying for regulations, patent trolling, occupational licensing cartels, corruption, feudalistic rent extraction, any activity focused on redistribution rather than value creation.
- **Steps**: (1) Distinguish between value-creating and rent-seeking activities in your strategy. (2) Calculate the full social cost of lobbying vs. innovating. (3) Identify rent-seeking by competitors and design regulation to block it. (4) Avoid engaging in rent-seeking even when profitable, it invites retaliation and regulatory scrutiny. (5) Advocate for institutional rules that reward production over position.
- **Examples**: India's "Licence Raj" lobbying; patent trolls; occupational licensing cartels; corruption and bribery as market entry barriers.

Market Design and Mechanisms

Mechanism Design: Engineering the Rules

- **Introduction**: Reverse game theory, instead of predicting what players do given rules, we ask what rules to design to elicit desired behaviour. It is the architecture of incentives, making truth-telling the best strategy.
- **Situation**: School choice systems, spectrum auctions, kidney exchange, tax code design, medical residency matching, any context where the designer controls the rules but not the players.
- **Steps**: (1) Define the social objective (efficiency, fairness, revenue). (2) Identify the private information players hold. (3) Design a mechanism where revealing true information maximises individual payoffs (incentive-compatibility). (4) Ensure participation is individually rational. (5) Test for manipulation and iterate.
- **Examples**: NRMP medical residency matching; FCC spectrum auctions; school choice algorithms; kidney exchange programmes; Vickrey second-price auctions.

Auction Theory: Strategic Bidding

- **Introduction**: Different auction formats (English, Dutch, sealed-bid, Vickrey) elicit different bidder behaviours and revenue outcomes. The winner's curse, overpaying because winning means everyone else bid lower, is the central hazard.
- **Situation**: IPL player auctions, government spectrum auctions, estate sales, eBay bidding, Google Ads, procurement auctions, privatisation of state assets.
- **Steps**: (1) Identify the auction format and its incentive structure. (2) Estimate your true private value, resist the winner's curse by bidding below your estimate in common-value auctions. (3) Observe competitor bidding patterns. (4) Use the Revenue Equivalence Theorem to evaluate format choices. (5) Bid truthfully in second-price (Vickrey) auctions, it is the dominant strategy.
- **Examples**: IPL team auctions for player talent; Google AdSense second-price mechanism; eBay online bidding; government bond issuance.

Matching Markets: Stable Allocations

- **Introduction**: Some markets cannot use prices to clear allocation (kidney donation, medical residency). Matching algorithms create stable outcomes where no unmatched pair would both prefer each other, preventing market unravelling and black markets.
- **Situation**: Medical residency assignment, school choice, kidney exchange networks, entry-level labour markets, organ donation registries, college admissions.
- **Steps**: (1) Verify that prices cannot or should not clear the market. (2) Gather complete preference lists from both sides. (3) Apply the Gale-Shapley deferred-acceptance algorithm. (4) Check that the resulting match is stable (no blocking pairs). (5) Design incentives so participants reveal true preferences.
- **Examples**: NRMP medical residency (Nobel Prize-winning application); school choice in NYC and Boston; kidney exchange chains; entry-level lawyer placement.

Network Effects: Platforms and Lock-In

- **Introduction**: The value of a product increases with the number of its users. This creates winner-take-most dynamics where leading platforms grow larger while challengers die, not because the product is better but because the network is bigger.
- **Situation**: Social media, payment networks, operating systems, languages, stock exchanges, communication platforms, any product whose value scales with user count.
- **Steps**: (1) Measure same-side and cross-side network effects. (2) Subsidise early adoption to reach critical mass. (3) Design for interoperability or, strategically, against it. (4) Use data network effects to reinforce technical advantages. (5) Identify multi-homing costs, high multi-homing weakens lock-in.
- **Examples**: WhatsApp's dominance despite imperfect product; Visa/Mastercard duopoly; Windows OS market; English as global business language.

Two-Sided Markets: Balancing Multiple Sides

- **Introduction**: Platforms facilitating interaction between two distinct groups face a chicken-and-egg bootstrapping problem. The strategy requires subsidising one side to attract the other, then extracting value from whichever side has inelastic demand.
- **Situation**: Ride-sharing, dating apps, gaming consoles, newspapers, credit cards, shopping malls, any platform requiring both supply and demand side to create value.
- **Steps**: (1) Identify which side is the "hard side" to attract. (2) Subsidise the hard side heavily to build critical mass. (3) Charge the side with the most to gain from the other's presence. (4) Manage cross-side network effects, more of one side should benefit the other. (5) Prevent disintermediation, keep both sides transacting on the platform.
- **Examples**: Uber subsidising drivers and riders; gaming consoles subsidising hardware to attract developers; newspapers giving free content to attract advertisers.

Collective Challenges and Social Dilemmas

Tragedy of the Commons: Overuse Destroys Resources

- **Introduction**: When a shared resource is unregulated, individuals consume it to exhaustion, the benefit of consumption is private, but the cost of depletion is borne collectively. Rational individuals destroy the commons.
- **Situation**: Overfishing, groundwater depletion, traffic congestion, air pollution, office kitchen misuse, climate change, any shared resource without governance.
- **Steps**: (1) Make the resource excludable (property rights or licensing). (2) Introduce quotas or Pigouvian taxes on usage. (3) Build community norms and social monitoring. (4) Apply Ostrom's principles: local rules, graduated sanctions, collective governance. (5) Create long-term stakes in the resource's health.
- **Examples**: Delhi's smog from uncurbed driving; global overfishing; groundwater depletion in agricultural regions; climate change as a global commons failure.

Voting Paradoxes: When Democracy Fails

- **Introduction**: Arrow's Impossibility Theorem proves that no rank-order voting system can perfectly satisfy all criteria of fairness simultaneously. Every voting system can be gamed or produce inconsistent collective choices.
- **Situation**: Political elections with multiple candidates, boardroom decisions, committee voting, Oscar nominations, budget prioritisation, any collective choice aggregating individual preferences.
- **Steps**: (1) Understand the voting mechanism and its incentives. (2) Identify potential spoiler effects and strategic voting incentives. (3) Consider alternative mechanisms (ranked-choice, approval voting). (4) Account for agenda-setting power, who controls the order of votes. (5) Design voting systems to minimise strategic manipulation given the context.
- **Examples**: Spoiler candidates splitting the vote; gerrymandering; strategic "lesser-evil" voting; Condorcet cycles in committee decisions.

Volunteer's Dilemma: Who Acts First?

- **Introduction**: When a group needs one person to incur a cost for everyone's benefit, each member waits for another to volunteer. The larger the group, the more severe the problem, resulting in paralysis even when action is clearly needed.
- **Situation**: Emergency bystander responses, whistleblowing, group project initiation, social movements, bug reporting, standing ovations, reporting organisational wrongdoing.
- **Steps**: (1) Assign explicit responsibility to a specific person, diffused responsibility equals paralysis. (2) Reward first-movers visibly and publicly. (3) Reduce the cost of volunteering (anonymity, indemnification). (4) Break large groups into smaller accountable units. (5) Build a culture where volunteering is the expected norm.
- **Examples**: Bystander effect in the Kitty Genovese case; whistleblowers in corporations; group projects where no one starts; first protesters in social movements.

Focal Points (Schelling Points): Coordinating Without Communication

- **Introduction**: When multiple equilibria exist and communication is impossible, people converge on solutions that stand out through prominence, symmetry, or cultural salience. Self-fulfilling expectations create convergence without explicit agreement.
- **Situation**: Emergency rendezvous, social conventions, negotiation anchors, ceasefire boundary selection, standard pricing points, any coordination without the ability to communicate.
- **Steps**: (1) Identify which options stand out as "obvious" in context. (2) Use round numbers and symmetric solutions as coordination anchors. (3) In design, make your desired outcome the salient focal point. (4) Use cultural and historical reference points to guide coordination. (5) Recognise that your opponents will also search for the same focal point.
- **Examples**: Grand Central Station at noon as a meeting point; round-number salary anchors (Rs. 10 lakh, not Rs. 9.83 lakh); ceasefire lines at rivers; splitting bills equally.

The Art of War and Diplomacy: Ancient Strategic Wisdom

- **Introduction**: Chanakya and Sun Tzu wrote millennia ago, yet CEOs quote them today. Because technology changes but human nature and conflict do not. The highest strategy is to win without fighting.
- **Situation**: Corporate competition, geopolitical diplomacy, organisational politics, competitive intelligence, brand wars, wherever perception management and asymmetric information can substitute for direct confrontation.

- **Steps**: (1) Know yourself and your enemy, assess strengths and weaknesses honestly. (2) Win without fighting: use perception, reputation, and deception before force. (3) Choose the battlefield, fight where you are strong, not where the enemy is. (4) Use spies and intelligence before commitment. (5) Strike at the opponent's strategy, not their army.
- **Examples**: Corporate hostile takeover via proxy; soft power (Bollywood/Hollywood); guerrilla marketing; peace treaties as strategic positioning; cyber warfare.

Prospect Theory

Prospect Theory: Loss Aversion Changes Everything

- **Introduction**: Kahneman and Tversky's theory shows humans evaluate outcomes relative to a reference point, not absolute wealth, and losses hurt roughly twice as much as equivalent gains feel good. This reshapes all strategic interaction.
- **Situation**: Investor behaviour, labour negotiations, political framing, consumer pricing, legal settlements, any context where framing around a reference point shifts decisions away from rational maximisation.
- **Steps**: (1) Identify the reference point of your counterpart. (2) Frame your offer as avoiding a loss rather than achieving a gain. (3) Recognise the endowment effect, people overvalue what they already possess. (4) Avoid triggering loss aversion by anchoring to a favourable reference point early. (5) Exploit loss aversion in competitors while designing your own decisions around rational expected value.
- **Examples**: Investors refusing to sell losing stocks; homeowners overpricing due to emotional attachment; unions rejecting wage cuts even to prevent layoffs; governments escalating losing wars to "not lose face."

Conclusion and Next Steps

The Infinite Game: Key Takeaways

What We Learned

- You are never playing solitaire, always chess
- Incentives trump intentions every time
- Information is the most valuable commodity
- Credibility must be earned through action
- Cooperation requires design, not hope
- Reputation compounds over decades
- The framing of a choice changes the choice

The Hierarchy of Strategic Wisdom

1. Know which game you are in
2. Know who the other players are
3. Know their incentives
4. Know your BATNA
5. Make credible commitments
6. Build reputation relentlessly
7. **Play to keep playing**

The Ultimate Lesson
In the *Finite Game* of a football match, you play to win. In the *Infinite Game* of life, you play to **keep playing**. Stay adaptable.

Applying Game Theory: Your Action Plan

This Week

- Map one negotiation you are in as a game tree
- Identify your BATNA and improve it
- Find one Nash Equilibrium you are stuck in and ask: *what changes the game?*
- Spot one dominant strategy and use it

This Month

- Redesign one professional relationship using Tit-for-Tat
- Audit your reputation signals
- Find one rent-seeking activity and eliminate it
- Implement one pre-commitment device for a personal goal

Applying Game Theory: Your Action Plan

Ongoing Practice

- Before every significant decision: *who else is deciding, and what are their payoffs?*
- In every negotiation: *what is the other party's reference point?*
- In every conflict: *is this zero-sum or can I grow the pie?*
- In every commitment: *is this threat or promise actually credible?*

Remember
Game Theory is a map, not the territory. Real people are irrational, emotional, and surprising. Use these models as lenses, not algorithms.

Essential Reading: Resources to Go Deeper

Foundational Books

- **Thinking Strategically**, Dixit & Nalebuff (best accessible intro)
- **The Art of Strategy**, Dixit & Nalebuff (applied game theory)
- **Games of Strategy**, Dixit, Skeath & McAdams (textbook)
- **The Strategy of Conflict**, Thomas Schelling (focal points, brinkmanship)
- **Thinking, Fast and Slow**, Daniel Kahneman (Prospect Theory)
- **Getting to Yes**, Fisher & Ury (applied negotiation)

Advanced & Specialised

- **Arthashastra**, Kautilya (Chanakya), ancient strategy
- **The Art of War**, Sun Tzu, conflict & deception
- **Predictably Irrational**, Dan Ariely, behavioural economics
- **Co-opetition**, Brandenburger & Nalebuff, business strategy
- **The Infinite Game**, Simon Sinek, finite vs. infinite perspective
- **Who Gets What and Why**, Alvin Roth, matching markets (Nobel)

Closing Reflection

Thinking Strategically . . .

- "The supreme art of war is to subdue the enemy without fighting." - Sun Tzu, The Art of War
- "In the real world, the game isn't broken. The players are human." - Kahneman / Tversky, Prospect Theory Reflection
- Understand the incentives.
- Respect the players.
- May your equilibrium always be stable.

CopyLeft ◎ 2014 Yogesh Kulkarni yogeshkulkarni@yahoo.com